

PD Data Collection

Android Application

Status Report & Data Samples

Date: February 15, 2026

Target: Android 10+ (API 29+)

Test Device: Samsung Galaxy S24 Ultra (SM-S928B)

This report summarizes the current state of the Parkinson's Disease data collection Android application. The app continuously collects sensor data (accelerometer, gyroscope, magnetometer), touch/keystroke interactions via Accessibility Service, periodic screenshots via MediaProjection, and administers the Trail Making Test (Parts A & B) twice daily. All data is stored locally on the device in CSV format.

Data Collected (test session, ~20 minutes):

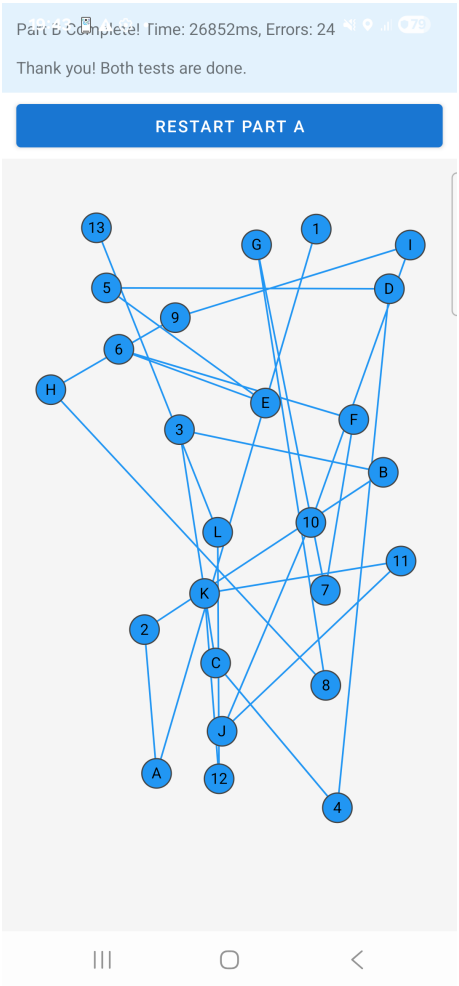
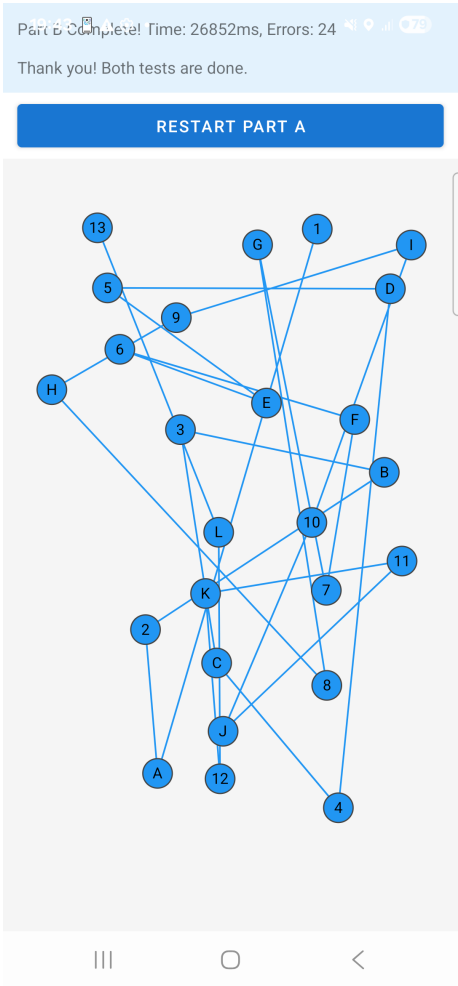
File	Rows	Size	Description
sensors.csv	324,016 rows	35 MB	Accel/Gyro/Mag at ~100Hz
touch.csv	30 rows	1.3 KB	Clicks and scrolls with coordinates
apps.csv	25 rows	976 B	App open/close events
tmt_results.csv	2 rows	4.1 KB	TMT Part A & B with full path data
screenshots/	9 files	344 KB	Periodic screen captures (JPEG)

1. Application Screens

The app follows a linear onboarding flow: Consent -> Profile Setup -> Main Dashboard. From the dashboard, users can start/stop data collection, run the Trail Making Test, edit their profile, or access settings.

Main Dashboard

The main screen shows the participant's profile summary, total data collected, and storage path. Buttons allow starting sensor collection, screen capture, and the Trail Making Test.

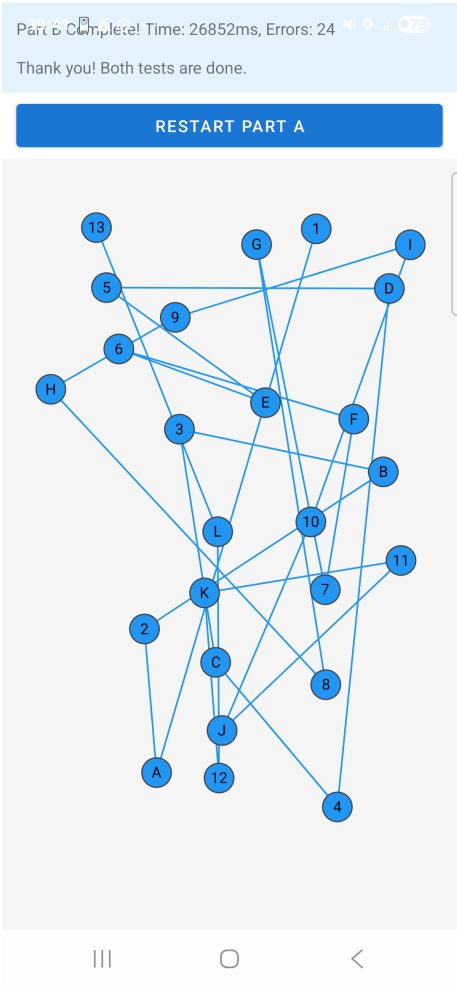
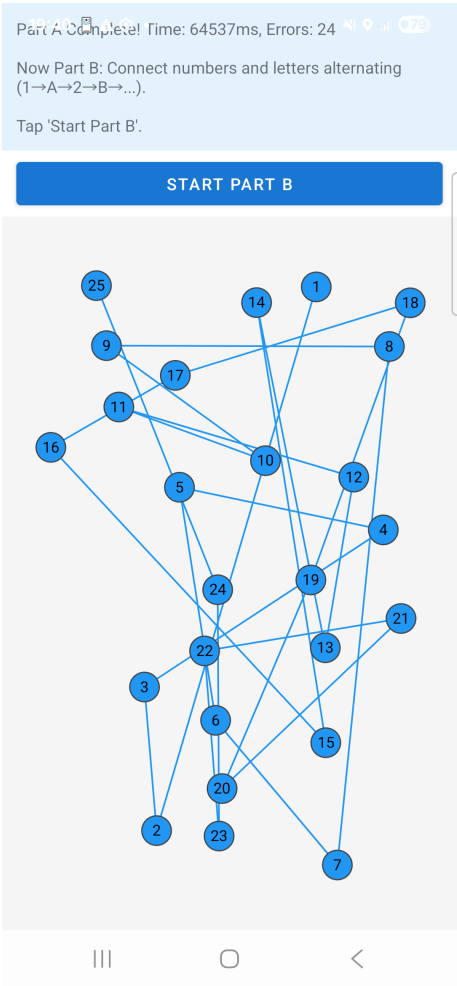


Main Dashboard

Settings Screen

2. Trail Making Test

The Trail Making Test (TMT) is a neuropsychological test of visual attention and task switching. Part A requires connecting numbers 1-25 in order. Part B requires alternating between numbers and letters (1-A-2-B-3-C...). The app records total time, errors, and the complete touch path with timestamps for each connection.



Part A Complete (1->25)

Part B Complete (1->A->2->B...->13)

Results from test session:

Part A: Total time = 64,537 ms, Errors = 24

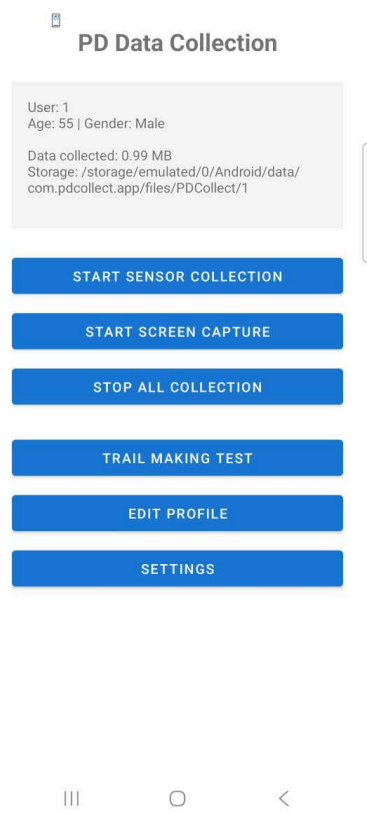
Part B: Total time = 26,852 ms, Errors = 24

Note: High error counts are due to automated (adb) tapping during testing. Real user finger taps will produce more accurate error counts.

3. Screen Capture

The app captures screenshots every 10 seconds using the MediaProjection API. A delta comparison algorithm only saves images when significant screen changes are detected (>5% pixel difference), reducing storage usage. Images are saved as compressed JPEG (quality 50).

Sample captured screenshot:



4. Data Format - CSV Samples

sensors.csv - IMU Sensor Data (~100Hz)

Columns: timestamp_ns, accel_x/y/z, gyro_x/y/z, mag_x/y/z

```
timestamp_ns,accel_x,accel_y,accel_z,gyro_x,gyro_y,gyro_z,mag_x,mag_y,mag_z
353387372959579,-0.047856454,0.37328035,9.427721,6.108648E-4,0.0018325944,0.0010690134,0.0,0.0,0.0
353387375092964,-0.033499517,0.3493521,9.523434,6.108648E-4,0.0018325944,0.0010690134,0.0,0.0,0.0
353387377226349,-0.023928227,0.3589234,9.5760765,-0.076358095,-0.09346231,0.0035124724,0.0,0.0,0.0
353387379361089,-0.023928227,0.33020952,9.676575,-0.076358095,-0.09346231,0.0035124724,0.0,0.0,0.0
353387381494683,0.0,0.32542387,9.79143,-0.076358095,-0.09346231,0.0035124724,0.0,0.0,0.0
```

touch.csv - Touch & Scroll Events

Columns: timestamp_ms, event_type, x, y, source_app

```
timestamp_ms,event_type,x,y,source_app
1771176234423,click,405,557,com.android.chrome
1771176236357,scroll,0,0,com.android.chrome
1771176246526,scroll,0,0,com.samsung.android.honeyboard
1771176248331,scroll,0,0,com.android.chrome
1771176252237,scroll,0,0,com.samsung.android.honeyboard
1771176253050,scroll,0,0,com.samsung.android.honeyboard
1771176255917,scroll,0,0,com.android.chrome
1771176257318,scroll,0,0,com.android.chrome
1771176276139,scroll,0,0,com.android.chrome
1771176304091,scroll,0,0,com.android.chrome
1771176313688,scroll,0,0,com.android.chrome
1771176346422,click,540,781,com.pdcollect.app
1771176368642,click,0,0,com.pdcollect.app
1771176388455,click,540,199,com.pdcollect.app
1771176745784,click,0,0,com.pdcollect.app
1771176746851,click,540,199,com.pdcollect.app
1771176769296,click,540,199,com.pdcollect.app
1771176916551,click,0,0,com.pdcollect.app
1771176917644,click,540,199,com.pdcollect.app
1771176938423,click,540,199,com.pdcollect.app
1771177037416,click,540,199,com.pdcollect.app
1771177062181,click,0,0,com.pdcollect.app
1771177063068,click,0,0,com.android.systemui
1771177063185,scroll,0,0,com.android.systemui
1771177065502,click,0,0,com.pdcollect.app
1771177068331,click,0,0,com.pdcollect.app
1771177129767,click,0,0,com.pdcollect.app
1771177131007,click,540,199,com.pdcollect.app
1771177248713,click,540,199,com.pdcollect.app
```

apps.csv - App Usage Events

Columns: timestamp_ms, event_type, package_name, class_name

```
timestamp_ms,event_type,package_name,class_name
1771176234708,open,com.samsung.android.honeyboard,android.inputmethodservice.SoftInputWindow
1771176313599,close,com.samsung.android.honeyboard,
1771176313599,open,com.pdcollect.app,com.pdcollect.app.ui.MainActivity
1771176739320,open,com.pdcollect.app,com.pdcollect.app.ui.MainActivity
1771176903700,open,com.pdcollect.app,com.pdcollect.app.ui.MainActivity
1771177062235,close,com.pdcollect.app,
1771177062235,open,com.android.systemui,com.android.systemui.statusbar.phone.AlertDialogWithDelegate
1771177065070,close,com.android.systemui,
1771177065070,open,com.pdcollect.app,com.pdcollect.app.ui.MainActivity
1771177065558,close,com.pdcollect.app,
1771177065558,open,com.android.systemui,com.android.systemui.statusbar.phone.AlertDialogWithDelegate
1771177066417,close,com.android.systemui,
1771177066417,open,com.pdcollect.app,com.pdcollect.app.ui.MainActivity
1771177128732,open,com.pdcollect.app,android.widget.FrameLayout
```

tmt_results.csv - Trail Making Test Results

Columns: timestamp_ms, test_type, total_time_ms, errors, path_data_json

The path_data_json field contains a JSON array of touch events with:

t = time offset from test start (ms)

x, y = touch coordinates

a = action type (0=DOWN, 1=UP, 2=MOVE)

```
timestamp_ms, test_type, total_time_ms, errors, path_data_json
```

```
1771177195489, A, 64537, 24, <2079 chars of path JSON>
```

```
1771177275507, B, 26852, 24, <2079 chars of path JSON>
```


5. Architecture Overview

The application consists of the following components:

ConsentActivity

First screen. Displays research data collection warning and requires explicit consent checkbox before proceeding.

ProfileSetupActivity

Collects participant ID, age, gender, and medications (name + dose). Stored in SharedPreferences.

MainActivity

Central dashboard showing collection status, data size, and controls to start/stop services and launch the TMT.

SensorCollectionService

Foreground service registering accelerometer, gyroscope, and magnetometer at `SENSOR_DELAY_FASTEST`. Buffers readings and flushes to CSV every 5 seconds.

DataAccessibilityService

Accessibility service capturing clicks (with coordinates), text changes (keystroke timing), scroll events, and app switches (foreground package changes).

ScreenCaptureService

Uses MediaProjection API to capture screenshots every 10 seconds. Compares with previous frame and only saves if >5% pixels changed. Saves as compressed JPEG.

TrailMakingTestActivity

Interactive canvas-based TMT implementation. Part A: numbers 1-25. Part B: alternating numbers and letters. Records full touch path with timing.

BootReceiver

Automatically starts data collection services after device reboot if consent was previously given.

TMTReminderReceiver

Schedules twice-daily alarms (10:00 and 16:00) to remind participants to complete the Trail Making Test.

DataManager

Central file I/O manager. Creates date-based directory structure, manages buffered CSV writers, handles file rotation at midnight.

6. Storage Structure

All data is stored on the device at:

```
/sdcard/Android/data/com.pdcollect.app/files/PDCollect/<userID>/<date>/
```

Directory layout:

```
PDCollect/
  1/                                # User ID
    2026-02-15/                     # Date
      sensors.csv                   # IMU data (accel, gyro, mag)
      touch.csv                    # Touch/scroll events
      keys.csv                      # Keystroke events
      apps.csv                     # App open/close events
      tmt_results.csv              # Trail Making Test results
      screenshots/                 # Periodic screen captures
        screen_<timestamp>.jpg
```

7. Permissions Required

The app requires the following permissions:

- * `FOREGROUND_SERVICE` - For continuous sensor collection
- * `FOREGROUND_SERVICE_SPECIAL_USE` - For research data collection service
- * `FOREGROUND_SERVICE_MEDIA_PROJECTION` - For screen capture
- * `POST_NOTIFICATIONS` - For status notifications and TMT reminders
- * `RECEIVE_BOOT_COMPLETED` - To auto-start after reboot
- * `SCHEDULE_EXACT_ALARM` - For TMT reminder scheduling
- * `HIGH_SAMPLING_RATE_SENSORS` - For >100Hz sensor data
- * `REQUEST_IGNORE_BATTERY_OPTIMIZATIONS` - To prevent Doze mode interruption
- * Accessibility Service - Must be manually enabled in Settings
- * MediaProjection - User must grant screen recording permission

8. Next Steps

1. Deploy to test devices (5-10 PD patients)
2. Configure TMT reminder times per participant schedule
3. Set up periodic data extraction (USB or network sync)
4. Monitor battery impact and adjust sensor sampling if needed
5. Add data encryption for sensitive keystroke/screenshot data
6. Consider adding medication timing prompts